Evidence Selection for GTM Research Plays: A Controlled Study of Catalog-Conditioned Memory

Technical report prepared for Pancake

22 October 2026

Abstract

Company memory can contain the facts needed for an action without enabling a model to use them correctly. We study this distinction through a controlled GTM research-Play task: recommend a catalog workflow with current supporting evidence, or withhold when its prerequisites are unsatisfied. Six evidence-selection adapters share verified fictional records, a catalog, a quantized Qwen3 reader and a constrained output interface. The frozen study comprises 12 company worlds, 96 queries, 576 primary comparisons and 192 larger-budget comparisons, executed through 676 actual model requests with 92 identical-request reuses. At a 1,024-token input allowance, catalog-conditioned coverage achieves 53/96 correct supported decisions versus 40/96 for prompt-specialized hybrid retrieval. The paired difference is 13.5 percentage points, with a family-stratified world-bootstrap interval of [0.0, 28.1]; the primary comparison does not establish clear superiority. Catalog coverage retrieves complete support for all 48 answerable cases, but the reader produces only 21 supported recommendations and recommends inappropriately on 16/48 negative cases. A 1,536-token exploratory comparison favors catalog coverage, while an exact SQL reference solves every formal case. These results support prerequisite-aware evidence selection as a component to investigate, but expose substantial utilization and constraint-checking failures. They do not establish strategic usefulness, raw-document ingestion quality, production readiness or commercial advantage.

1 Introduction

Company memory is useful only if a decision can use the right evidence. In go-to-market (GTM) work, a seller may retain its approved target market, buyer roles, product claims and proof, while separate records describe prospective accounts and previous actions. A plausible recommendation can still be unusable: the buyer role may be wrong, the supporting signal may have been corrected, or the evidence may be unavailable to the requester. Conversely, refusing every action avoids many unsupported proposals while providing little useful coverage.

We study a limited interface between these records and action: choosing a research *Play* from a shared catalog, or requesting evidence when no catalog alternative is supported. A Play is a scoped workflow with explicit prerequisites and a resource allowance. The interface emits a proposal and its sources; a deterministic validator can materialize an eligible proposal into a specification and execute its research steps against fictional providers. This establishes consistency with an explicit contract. It does not establish that a Play is commercially optimal or will improve revenue.

The motivating Pancake product direction already treats its Brain as approved GTM context for Plays. We investigate which evidence-selection mechanisms help within that domain, using a standalone prototype and fictional records. The experiment does not measure the production Brain or introduce its first domain semantics.

The motivating question is whether task-specific memory structure improves this interface beyond capable generic retrieval and a domain prompt. The question is easy to confound. A specialized system may receive cleaner facts, a larger context, an extra model call, or a better planner. A generic system may be handicapped by missing domain vocabulary. High retrieval recall can also coexist with incorrect recommendations. We therefore hold the verified facts, metadata boundary, catalog, reader and output interface constant, and compare retrieval selection under an explicit total-input allowance. We include a prompt-only specialization control, a domain-neutral diversity control, and separately priced full context.

The contribution is a controlled empirical study and an executable fixture suite. It is not a new memory architecture. Marketing memory and planning, domain-aware memory, belief-based recommendation, complementary evidence retrieval and derived-state maintenance all have close precedents. Our narrower question concerns catalog-conditioned evidence coverage for GTM research proposals under changing policy, observations and permissions. Because the selection mechanism can operate on any catalog of typed prerequisites, a benefit would not establish a uniquely GTM algorithm or an ontology-based commercial advantage.

We distinguish three layers throughout. A *system of record* preserves versioned evidence and approved policy. *Decision support* proposes a Play with complete relevant support. An *action interface* checks and dry-runs a specification. None alone proves strategic usefulness. The study evaluates the latter two layers conditional on clean, structured records; it does not evaluate extraction from raw documents, production deployment, cross-company learning or campaign outcomes.

2 Related work and scope of the review

The review used primary papers and author-maintained artifacts available by 22 October 2026. Ten first-page arXiv searches spanning agent memory, domain specialization, temporal knowledge, recommendation, marketing, sales, evaluation and classical foundations yielded 399 distinct records. Automated title triage, targeted citation tracing and deeper reading of the closest papers are logged separately. These are bounded searches, not an exhaustive systematic review; many candidates and appendices remain unread. Canonical bibliographic metadata and source hashes are retained in the companion evidence, with review depth recorded independently of inclusion in the bibliography.

Marketing and recommendation already use memory. RAMP combines client-specific semantic and episodic memory with marketing planning, executable audience filters, verification and reflection [1]. Its joint planner/memory intervention and small human study address a richer audience-creation workflow than ours; they do not isolate memory representation under a matched total budget. Its repair behavior can relax audience criteria, whereas our contract treats approved constraints as fixed until an explicit revision. MARS connects hierarchical belief-state memory to recommendation [2]. This directly rules out novelty claims for evolving company knowledge leading to recommendations. We study explicit support and eligibility, not a learned recommendation utility or calibrated belief confidence.

Domain memory is a task-dependent question. IFCMemoryBench connects prior conversations to live structured building-model queries [3]. Its separation of ingestion, retrieval and utilization, and its sensitivity to the probe agent, motivate our common-reader boundary. AFTER evaluates procedural memory across enterprise tasks and roles [4]; reusable procedures and transferable knowledge should not be assumed equivalent. SalesResearch and CRMArena-Pro evaluate sales-oriented agent work [5, 6], but different task interfaces, backbones and simulation assumptions

prevent using their scores as a memory-only baseline for this study. Our fictional catalogs measure neither those systems' capabilities nor production sales performance.

Temporal evidence and maintenance have close precedents. LongMemEval and its later evaluation work make temporal reasoning, updates, abstention and contextual baselines central [7, 8]. Zep uses temporal graph memory [9]; Hindsight separates factual, experiential and belief-oriented operations [10]. RECON examines consequences of changed evidence [11]; AkasicMEM proposes lineage and authorization continuity [12]. Most directly, EviGraph connects temporal evidence chains to supported decisions, and MemDerive tests derived-state correctness and maintenance [13, 14]. We do not claim to introduce temporal provenance, proof-supported recommendation or dependency tracking. Our simple eligibility view is a shared experimental boundary, not a replacement for these complete systems or a security proof.

Strong simple retrieval changes the comparison. MemCompile emphasizes complementary evidence packets rather than independently relevant passages [15]. Controlled comparisons of memory systems with raw-history retrieval show why reader strength, retriever strength and context allowances must be separated [16]. ReFind offers another relevant retrieval-oriented comparison [17]. These works motivate lexical–dense fusion, a generic relation-diversity control, explicit input-token measurement and full context. Standard BM25 and reciprocal-rank fusion have established foundations [18, 19]; using them here is implementation reuse of standard methods, not an algorithmic contribution. We do not benchmark deployed Hindsight, Graphiti or other full ingestion systems, because doing so would introduce additional extraction and maintenance differences. Conclusions are restricted to the implemented adapters.

MemoryArena explicitly evaluates interdependent multi-session actions and memory updates from environment feedback [20]. Its verified ICML 2026 proceedings and controlled task construction make the memory-to-action overlap clear. Our reader is stateless between queries and receives already verified records; successful proposals here cannot establish the closed-loop capabilities evaluated there.

Mem0 evaluates extracted and maintained conversational facts, with a graph variant and a full-context reference [21]. Its graph and base configurations have different retrieved-token budgets, and full context retains the highest overall judge score in the reviewed paper; such results motivate conditional quality/cost comparisons rather than an assumed graph advantage. Anti-Persistence examines a complementary risk: inferred guidance can influence later trajectories and abstractions [22]. Its qualified observational analysis does not prove inevitable error growth, but it reinforces the distinction between observed records and inferred policy. Our static-history experiment cannot test that learning loop.

The surviving distinction is consequently modest: an auditable GTM research-Play contract, a shared verified-event boundary, and a paired comparison separating domain instructions, generic evidence diversity and public-catalog prerequisite coverage. The study can reveal an engineering boundary or a negative result without establishing state-of-the-art agent memory.

3 Task and evidence boundary

Each query specifies a company world, target account, objective, valid-time cutoff, knowledge-time cutoff, requester permissions and maximum resource units. An event contains its source identifier, world, entity, relation, value, authority, valid time, observed time, required permissions and explicit supersession links. Authority distinguishes observed evidence, approved policy, proposals and

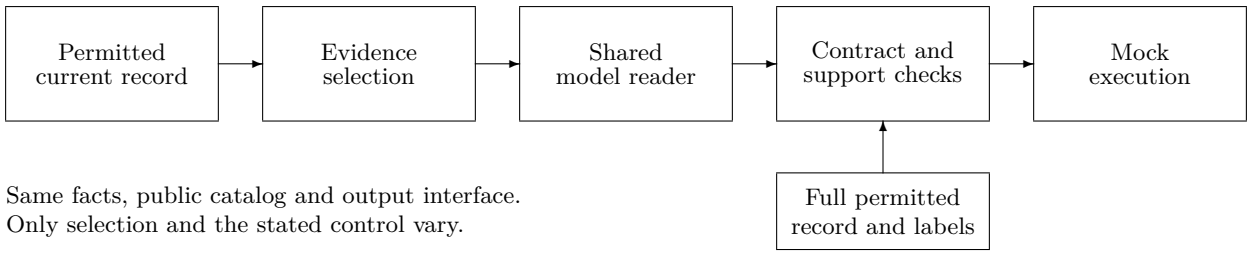


Figure 1: The controlled evidence boundary. The time/access view precedes every retrieval adapter. The reader sees selected evidence; evaluation also has the full permitted record and hidden labels. Labels never enter retrieval or the reader. The full-context reference skips bounded selection. A validated positive proposal alone reaches the fictional executor. Correct withholding is scored separately.

hypotheses. Seller strategy and account observations remain logically separate even though the prototype stores both in local files.

The common view admits an event only when it belongs to the query's world, both time cutoffs are satisfied, and all required permissions are present. An eligible later event can supersede an earlier source only for the same world, entity, relation and authority. Independent conflicting current values remain a conflict; a convenient latest value is not silently chosen. This is a deliberately limited event model with valid-from times and explicit replacements, not a complete implementation of arbitrary bitemporal intervals.

A catalog template has an objective, required entity/relation/authority slots, optional literal values, equality constraints, workflow steps and resource units. Four common slots require approved seller role/geography and observed account role/geography, with the two pairs matching. Task-specific slots connect account state or signals to an offering pain and proof. Permission to read a source is distinct from approval of its content. A single non-response is an observation and cannot change approved targeting.

For a concrete contract illustration, development world `w69bf7897` targets IT managers in the EU. Fictional account Velora-231 matches those attributes and has an observed hiring signal; separate seller records connect its offering and proof to onboarding. With the required permissions and three available units, either the three-step research review or the two-step evidence review is acceptable. A procurement-specific alternative additionally requires a source absent from that view. Removing proof access blocks the hiring proposals, while a qualification-only objective needs only targeting evidence. This is an illustration of the declared fixture rules, not a model result or evidence that hiring predicts a purchase.

Let $V(q)$ denote the complete permitted current view, and $A(q)$ the set of catalog templates whose constraints are satisfied by $V(q)$. Several alternatives may be acceptable; we do not impose an arbitrary unique commercial winner. The reader returns either one template and cited sources, or a request for evidence. A supported recommendation must select a member of $A(q)$, cite at least one current supplied source for every required slot, and include no irrelevant sources. Withholding is correct only when $A(q)$ is empty. Returning nothing on answerable cases is an error.

All methods see the same public catalog and constraint language. The reader first fills a prerequisite-to-source table, with one supplied source or null for every catalog relation, and then selects a template or withholds. Short labels bind exactly to immutable identifiers. The common grammar allows every supplied label in every row, including irrelevant evidence; it does not choose a correct source. Rows required by the selected template become its citations. Missing entries

are not filled, and swapped row bindings fail even if their union contains all necessary sources. Rows concerning other templates do not become extraneous citations for the chosen Play. Identifier existence and parts of schema validity are enforced by construction, so they cannot be interpreted as empirical gains from memory. Correct source binding, completeness and template applicability remain model decisions. No post-hoc citation repair is performed.

After validation, a successful proposal becomes a Play specification with its objective, account, policy versions, sources, allowed next step, mechanism hypothesis and computed resource estimate. Mock research capabilities produce saved receipts from the complete permitted fictional provider view. They cannot revise the model's answer or add citations to make it pass. Each executed step costs one artificial resource unit. These units check workflow consistency; they are not dollars, conversion forecasts or estimated revenue. Mechanisms are catalog-authored hypotheses, not measured causal effects. The reader is stateless between queries: longitudinal cases test decisions over changing records, not autonomous revision of a running Play or learned experience reuse.

4 Comparison design

We isolate evidence selection on a common file-based backend. All adapters receive the same verified events, easy seller/account entity scope, time/access boundary and catalog. Thus the experiment does not test whether introducing typed facts is better than unstructured prose: every control already has typed facts and provenance. It tests how to select them for a bounded reader input.

The hybrid retriever combines BM25 ($k_1 = 1.2$, $b = 0.75$) with cosine similarity of normalized Qwen3-Embedding-0.6B vectors [23]. Reciprocal-rank fusion uses a constant of 60 and equal contributions. The query includes the account, objective, and prerequisite vocabulary of every objective- and budget-compatible template for *all* methods. The specialized method is therefore not alone in receiving the domain vocabulary. Ties are broken by immutable source identifier.

Generic hybrid. Fused relevance ranking with the common task contract.

Prompt hybrid. The same ranking plus a GTM checklist covering role/geography fit, signal/state, pain, proof and approved policy. This is the primary control.

Facet hybrid. The same prompt and ranking, traversing distinct entity/relation/authority groups round-robin before taking further members of a group. This control tests domain-neutral evidence diversity.

Catalog coverage. Required groups for templates matching the objective and resource allowance come first. Within each priority tier, fused ranking determines the initial group order; subsequent rounds preserve that order. Selection never reads gold labels or selects required values from answers.

Full context. Every eligible event, with the same checklist. This is a larger-input reference whose tokens and latency are reported separately.

History ablation. Catalog coverage with deterministic supersession removal disabled. Replacement links and times remain visible to the reader; permissions and time cutoffs remain enforced. This isolates the value of resolving replacements before retrieval in this implementation.

For bounded adapters, events are considered in ranking order and admitted only if the complete serialized prompt remains within the common token cap. Later smaller events can still fit after a longer event is rejected. The cap includes system instructions, query, catalog and evidence. The

generic prompt is shorter, so it can admit more evidence within the same total allowance. Selected events are finally presented in immutable-ID order under a common rule. Selection still changes positions and short source labels, so content availability and position effects are not independently identified. Full context fails explicitly above its safety cap; it is never silently truncated.

All methods share one planned reader decision, decoding settings, output allowance and binding grammar. No automatic retries, additional verification model or answer repair favor one adapter. The validator and mock execution follow the same rules for every output. Source selection can make a contract impossible to answer correctly, but withholding on an answerable complete record remains an error: this measures the end-to-end consequence of missing evidence rather than rewarding retrieval omissions.

Within each main block, identical serialized requests reuse the first actual response, including incorrect or malformed responses. The complete request body preserves schema field order and decoding settings; origins and raw outputs are retained. Seven repeated-input groups in the independently executed pilot returned identical raw outputs. This supports the resource-saving design without proving universal numerical repeatability. Logical comparisons and physical generation requests are reported separately. Method-level token/time totals charge the observed origin call to each identical request; physical totals count it once. No development generation output is imported into the main blocks.

An exact SQLite policy reference also receives the full verified record. It evaluates the declared joins and returns one witness per required slot, then passes through the same Python scorer and mock execution. This reference establishes the capability of an explicit rule evaluator on a rule-defined task. It is not a learned agent, an ingestion benchmark or evidence of commercial optimality.

The local implementation records exact weight hashes, upstream revisions, runtime build, prompts, source selections, model usage and terminal status. The token count from the runtime's chat template/tokenizer is checked against returned usage for each call. Embedding cache receipts retain their original computation times, so a method does not receive a free-ingestion advantage from running after another method. Supplementary accounting deduplicates shared embedding origins across blocks and estimates each method's component cost by charging its unique origins, non-embedding retrieval overhead and logical generation work. These are estimates from shared execution, not measured isolated deployments. Embedding time is already included in observed retrieval time and is not added twice. Times are elapsed wall-clock measurements on the CPU server, not accumulated process CPU time. Local API charges are zero; elapsed time, hardware rental value and energy are different quantities, and the latter two are not measured.

5 Data and evaluation

The dataset contains original programmatically authored fictional company worlds. The research assistant authored the generator and text templates; instantiation uses no separate narrative-generation model calls. There are no customer records or human annotations. Construction rules specify acceptable alternatives; separately implemented Python and SQLite evaluators verify the time/access and contract logic. This is software cross-validation by the same research process, not independent authorship or expert validation.

Each world has eight queries: initial eligibility; weak feedback and a proposed, unapproved policy; approved policy revision; account requalification; a later-observed backdated correction; restricted proof access; a changed objective; and conflicting approved policy. Four queries permit a supported Play and four require withholding. Always withholding therefore receives 50% decision accuracy. The late-correction case holds the earlier valid-time cutoff while advancing knowledge

time, distinguishing the two temporal axes. The access and objective interventions query earlier snapshots under changed permissions or purpose; the eight conditions are not one uninterrupted chronological sequence.

Pilot worlds use hiring and migration contracts. The main set contains 12 worlds and 96 queries, with four worlds per held-out renewal, partner and compliance family. These add different relation combinations. Worlds vary names, buyer roles, geographies, date gaps, template-ID permutations, resource limits, distractors and duplicate evidence. Queries from a world remain together. All families share the same generator and common policy slots; family holdout is compositional generalization within this generator, not an independently sourced domain transfer test. Structural checks inspect all public contract definitions, but no main-model outcome is used for tuning.

The primary block compares all six methods at 1,024 total input tokens, with 576 planned logical attempts. An exploratory block compares prompt hybrid and catalog coverage at 1,536 tokens on the same queries, adding 192 planned attempts. Full context is separately costed. Twelve worlds follow a prospective precision rule using a paired-world SD floor of 0.20, because two pilot worlds are insufficient for stable variance estimation, and an approximate interval half-width target of 0.15. This heuristic does not guarantee achieved precision. The exact rule, pilot outcomes, code/data hashes and ordering seeds were recorded before main inference.

The primary outcome is a *correct supported decision*: a completely supported applicable recommendation, or correct withholding when no alternative is supported. Every planned attempt remains in its denominator; failed, malformed, truncated and missing attempts cannot disappear. Separately reported dimensions include applicable template choice without complete citations, positive-case supported coverage, available support in retrieved evidence, inappropriate withholding, recommendations on negative cases, schema errors, mock execution completion and computation. Free-text reasons for withholding are retained but are not scored for semantic correctness; a correct negative decision does not establish a correct explanation.

The primary contrast is catalog coverage minus prompt hybrid. Facet diversity, the generic prompt, full context and history handling are explanatory comparisons. Queries and methods are paired within company worlds. Uncertainty uses world-level differences and a bootstrap stratified by held-out family, preserving all repeated queries from a sampled world together. Confidence intervals describe variation within the fictional generator and tested model, not commercial-population uncertainty. A small or degenerate interval cannot certify external validity, and nonsignificance does not establish equivalence. The saved protocol specifies sample selection and all settings before main execution.

Transition diagnostics report the fraction of worlds correct both before and after stable feedback or an evidence/policy change, as well as the corresponding conditional rate among correct earlier decisions. They do not manufacture new independent samples from adjacent queries or prove dependency-aware incremental recomputation. Raw output and analysis artifacts permit recomputation of every reported count. The prospective freeze was recorded in the working repository before inference; it was not an external preregistration.

6 Results

Both frozen blocks completed, with all 768 planned logical attempts retained. The primary block issued 494 actual reader requests and reused 82 identical requests; the sensitivity block issued 182 and reused 10. No execution failure, output truncation, missing usage or schema-invalid response was recorded. Schema validity is partly enforced by the common grammar and is not evidence of recommendation correctness. Both provenance audits verified the saved inputs and requests.

6.1 The primary contrast is positive but uncertain

At 1,024 input tokens, catalog coverage produces 53/96 correct supported decisions (55.2%), compared with 40/96 (41.7%) for prompt hybrid. The paired world difference is 13.5 percentage points, with a 95% stratified bootstrap interval of [0.0, 28.1]. Because this interval includes zero, the primary comparison does not establish clear superiority. The observed paired-world SD is 0.284, above the 0.20 planning floor. The achieved interval must therefore be read as observed, rather than as a guaranteed planning precision.

Method	Decision	Positive	Neg. rec.	Tokens	Gen. sec.
Generic hybrid	50.0%	4.2%	4.2%	1015	37.5
Prompt hybrid	41.7%	0.0%	16.7%	1013	36.6
Facet hybrid	35.4%	0.0%	29.2%	1012	35.6
Catalog coverage	55.2%	43.8%	33.3%	1016	34.0
Full context	56.2%	52.1%	39.6%	2099	67.1
History ablation	31.2%	35.4%	72.9%	1016	33.5

Table 1: Primary results. Decision is correct supported decisions over all 96 queries; Positive is supported recommendations over 48 answerable queries; Neg. rec. is inappropriate recommendations over 48 negative queries. Tokens are mean input tokens per planned query; Gen. sec. is median generation-request elapsed time, charging the origin call for reused requests. Full context has its separate larger allowance.

The difference is not a uniform improvement in caution and coverage. Catalog coverage answers 21/48 positive cases correctly, compared with none for prompt hybrid, but makes 16/48 inappropriate negative recommendations versus 8/48. Across paired queries, catalog coverage alone is correct on 26, prompt hybrid alone on 13, both on 27, and neither on 30. Always withholding scores 48/96 on this balanced task. Generic hybrid also scores 48/96, through two supported positive recommendations and 46 correct negative decisions. Neither aggregate accuracy nor a conservative response pattern alone demonstrates useful coverage.

6.2 Evidence selection and evidence use fail separately

Catalog coverage and full context each supply a complete witness for all 48 positive cases. Prompt hybrid supplies one in only 2/48, and generic and facet hybrid in 5/48 each. The smaller input allowance can nevertheless hold a complete witness for every positive query, as the pre-results capacity audit establishes. These are selection failures under the implemented ranking and distractor distribution, not a proof that generic memory cannot represent the necessary facts.

Complete support does not solve utilization. Among catalog coverage's 48 positive cases, 21 succeed, 13 withhold despite complete support, 13 choose an inapplicable template, and one has invalid support bindings. Full context succeeds on 25, with six inappropriate withholds, 15 inapplicable choices and two binding failures. These descriptive categories do not isolate a causal effect of evidence availability from positions, aliases or other input changes. All supported positive decisions complete the mock workflow; this dependent consistency check adds no independent evidence of strategic usefulness.

Condition-level results expose a sharp boundary. At the primary cap, catalog coverage fails all 12 changed-objective queries despite retrieving their complete qualification support, while full context solves eight. In the first lexicographic withholding example, query `q0b2e9ee8ab`, primary attempt 167 correctly fills all four common qualification bindings but requests partner-specific facts that

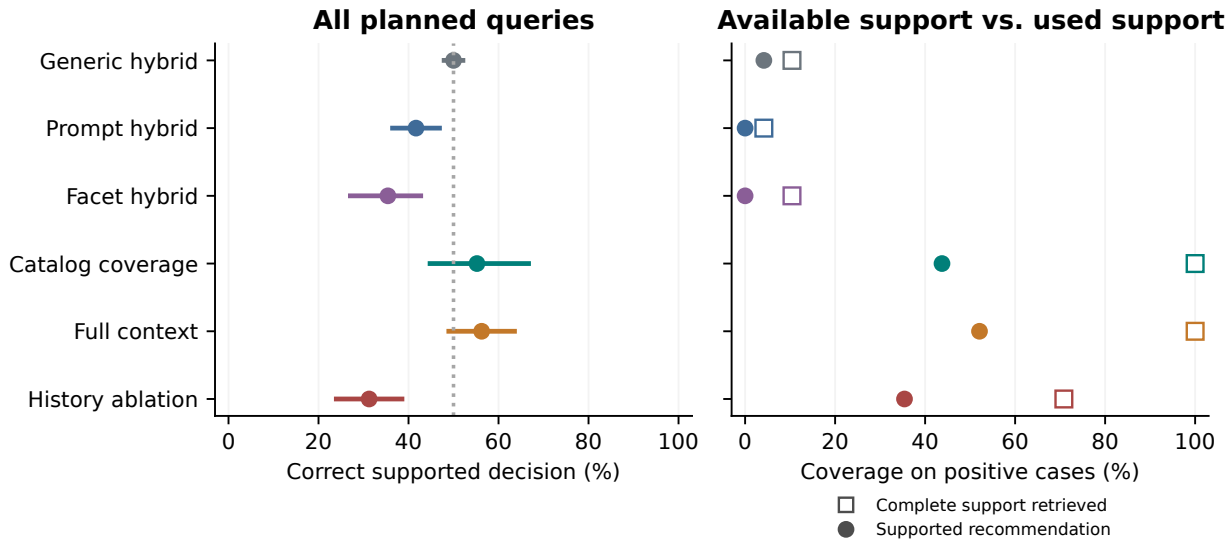


Figure 2: Primary decision quality and positive-case support. Left: method accuracy with world-bootstrap intervals; the dotted line is always withholding. Right: complete support retrieved versus correctly supported recommendations. Available evidence is necessary under this scorer but is plainly insufficient for this reader. Intervals describe resampling of the fictional worlds.

the qualification template does not require. Full-context attempt 211 selects the valid qualification template. This is a recorded alternative-scoping failure, not an explanation inferred solely from a score. Additional examples and all disagreements remain in the evidence package.

6.3 Controls reveal costs of instructions and unresolved history

Table 2 reports every frozen primary-block contrast. Catalog coverage exceeds facet hybrid descriptively, so generic diversity does not close the observed gap in this configuration. The prompt-specialized control scores below the generic control. This combines instruction effects with evidence displaced by the longer checklist under a total-token cap; it does not isolate the effect of wording alone.

Paired contrast	Difference (pp)	95% interval (pp)
Catalog coverage − prompt hybrid	13.5	[0.0, 28.1]
Catalog coverage − facet hybrid	19.8	[4.2, 38.5]
Prompt hybrid − generic hybrid	−8.3	[−15.6, −1.0]
Catalog coverage − history ablation	24.0	[17.7, 31.3]
Full context − catalog coverage	1.0	[−11.5, 13.5]

Table 2: World-paired effects at the primary cap. Only the first contrast is primary; the others explain behavior without a multiplicity-adjusted superiority claim. Full context is separately costed. An interval spanning zero does not establish equivalence.

Removing supersession resolution lowers catalog coverage from 53/96 to 30/96 decisions. The current-view method is correct alone on 23 paired queries; the history ablation is never correct alone. The ablation cites obsolete or ineligible evidence on 32 attempts and recommends on 35/48 negative cases. Its positive support availability also falls to 34/48, so this comparison includes

stale-evidence crowding as well as interpretation of replacement links. It supports resolving the explicit replacement semantics used here, not a claim of general learned belief revision.

Transition behavior remains incomplete. Catalog coverage is correct both initially and after weak feedback in 6/12 worlds (6/8 conditional on initial correctness). It is correct both before and after the approved policy revision in only 1/12 (1/6 conditional on the earlier decision). After account requalification and the later evidence correction, it is correct on both in 7/12 (7/7 conditional); the history ablation achieves 0/12. These paired snapshots do not test an agent updating a running Play or learning from its own experience.

6.4 Larger inputs help without establishing reliability

At 1,536 input tokens, catalog coverage scores 59/96 (61.5%) and prompt hybrid 42/96 (43.8%). The exploratory paired difference is 17.7 points [10.4, 26.0]. It does not replace the uncertain primary result. Positive support availability rises to 36/48 for prompt hybrid and stays at 48/48 for catalog coverage; supported positive recommendations are 13/48 and 30/48 respectively. Both methods recommend inappropriately on 19/48 negative cases.

Catalog coverage now solves 10/12 changed-objective queries, versus zero at the smaller allowance. Extra context changes more than the existence of a sufficient witness: its contents, positions and source labels also change. This improvement therefore cannot be attributed to missing prerequisites alone. Conversely, prompt hybrid gains 13 supported positive recommendations but loses 11 correct negative decisions relative to its primary-block result. More evidence is not a monotonic safety improvement.

6.5 Computation and the exact-rule reference

Mean primary input usage is 1,015.6 tokens for catalog coverage, 1,013.0 for prompt hybrid and 2,098.6 for full context. Their median generation times are 34.0, 36.6 and 67.1 seconds. Estimated component totals, charging each method its associated embedding origins and logical generation work, are 3,389.9, 3,605.8 and 6,557.5 seconds respectively. These shared-run estimates support a descriptive resource comparison, not a measured commercial lifecycle-cost advantage or a proof that the methods are equally accurate.

Across both blocks, 676 physical requests consume 882,141 input and 119,434 output tokens. Unique associated embedding work comprises 313 event and 24 query representations, taking 41.8 seconds in total. That work is already nested in observed retrieval time. The blocks consume 30,853.6 seconds (8.57 hours) of elapsed server time; local API charges are zero, while hardware value, energy, extraction and curation costs are unknown. Full computation receipts distinguish actual shared work from per-method charges.

The exact SQLite reference solves all 96 queries with zero generation calls and 0.084 seconds of recorded policy-evaluation time, excluding loading and other setup. That timing is not comparable to an end-to-end model deployment. Its correctness is nevertheless a substantive boundary: when the facts and eligibility language are already explicit, formal rule evaluation is sufficient. The learned-reader results must not be used to argue that an LLM is necessary for this contract.

7 Interpretation and product implications

The study yields a conditional engineering result rather than a validated intelligent GTM planner. Catalog-conditioned selection consistently makes a complete positive witness available under the tested small input budget. That component gain reaches some supported proposals, but many

reader failures remain, including recommendations that violate explicit policy or resource constraints. The primary accuracy interval includes zero, and the larger-budget result is exploratory. The evidence is insufficient for a broad claim that a specialized Brain produces superior commercial recommendations.

The central distinction is between storing evidence, selecting a sufficient set, and deciding correctly from it. Relevance ranking alone performs poorly on these deliberately repetitive distractors. Generic relation diversity also fails to recover complete witnesses reliably under the smaller allowance. Required-slot coverage is a plausible selection mechanism for an explicit catalog, but its algorithm is not uniquely GTM-specific. These results do not establish a graph advantage, a need for a new database, or superiority over full deployed memory systems.

For Pancake, the most direct implication is to preserve approved strategy and provenance, resolve explicit replacements, and check formal prerequisites outside the generative reader. The proposed Play should retain its policy versions, evidence bindings, objective and resource allowance. A validator should distinguish a supported proposal, a violated constraint and missing evidence before any real execution. The SQL result and the reader's negative-case errors support this division of responsibility on the tested clean-record task. They do not establish how much extraction or policy-authoring work is needed to make it reliable in practice.

The next research boundary is therefore evidence quality and decision usefulness under less favorable inputs. Independently authored histories should vary catalog alternatives more substantially; raw-document ingestion should charge extraction and maintenance cost; and stronger readers or different interfaces should be compared on a new frozen split. Blinded GTM assessment could then measure whether eligible proposals are useful, specific and worth a human's time. Campaign outcomes would require a separate prospective evaluation. Those are unmeasured questions, not implied benefits of the present scores.

Full context remains a serious reference at this record size. Its slightly higher primary point estimate comes with more input and generation time, but the paired interval is too broad to claim equivalence or a reliable quality ordering. A product should consequently choose between full context and selection using its actual context sizes, failure costs and reader, rather than treating this experiment as a universal architecture prescription.

8 Limits of the evidence

The experiment deliberately makes eligibility auditable, which also limits what its results mean. Its rules determine whether a research proposal has the required support. They do not establish that the rules capture a good commercial strategy, that one eligible Play is preferable to another, or that a signal predicts purchase intent. The two principal alternatives use the same prerequisites and differ mainly in workflow length. This favors a simple union of required relations; catalogs with heterogeneous alternatives could exceed the budget even when one alternative's support fits. The changed-objective task adds a simpler qualification check. This is a small selection problem, not open-ended campaign planning. A successful mock execution checks a fixed sequence against fictional providers; it measures neither provider reliability nor an outreach outcome. Because execution follows the same contract checks, its completion is largely a consistency diagnostic rather than independent evidence of usefulness.

Clean records and shared safeguards. Every adapter receives typed, verified facts and the same time/access filter. The experiment excludes extraction errors, entity resolution, approval workflows, storage migration and the labor of maintaining an ontology. These costs could outweigh

a retrieval advantage. Shared filtering and the constrained response grammar remove several opportunities for invalid access and invented identifiers. Their success is an implementation property, not evidence that one memory adapter learned authorization or is secure against adversarial content. Explicit supersession covers only the event semantics defined here; it is not general belief revision. The reader makes a fresh decision for each query and does not update a running Play or learn from experience.

Construction and external validity. The same AI-assisted research process authored the fixture generator, methods and scorer. Separately implemented Python and SQL checks detect software disagreements but cannot supply independent expert judgment about GTM realism. All test worlds come from one generator and three fixed contract families; held-out families still share policy relations and a common schema with development. Distractors deliberately repeat account/objective vocabulary without supplying required relations, stressing relevance versus complementary coverage; their frequency is not estimated from business documents. Pain and proof are categorical fixture relations, without an assessment of the strength of actual customer evidence. Randomized names, times and identifiers reduce trivial cues without creating independent business histories. The world bootstrap describes variation among these limited generated realizations, not a population of real companies. Distractor levels form a balanced stress grid, so the intervals should be read as resampling summaries, not a calibrated guarantee over arbitrary future histories. Neither the 96 queries nor repeated methods increase the count of independent worlds beyond 12.

Reader, budgets and tuning. The study uses one quantized, non-thinking reader on CPU, one embedding model and a compact prerequisite-binding interface. The 1,024- and 1,536-token allowances deliberately stress selection in relatively small records. A stronger reader, another serialization, longer inputs, tool-assisted verification or unstructured documents could change the ranking. Full context is a useful reference within these sizes, not a scalable treatment of arbitrary company history. The retained development sequence includes changes to readers, citation bindings and instructions; this effort was shared across methods before the main freeze, but it remains researcher choice. The study does not establish superiority to complete deployed memory products.

Measurement and reproducibility. The primary outcome combines positive support with correct withholding. A method can therefore gain accuracy by recognizing negative cases while still failing to serve positive ones; coverage and its components must accompany the aggregate. The free-text explanation of missing evidence is not semantically scored. No human assessment of strategic usefulness, calibrated confidence, revenue, retention or network effects is available. Identical-request reuse makes the main experiment conditional on one observed response per serialized request. Saved-output reproduction checks the reported analysis; fresh inference may vary with hardware, runtime or model availability. We measure tokens and elapsed time, not infrastructure dollars or energy. Other research and document-preparation processes shared the host, so latency is a descriptive deployment measurement rather than an isolated hardware benchmark. These limits preclude a claim of measured total economic advantage.

9 Conclusion

On a small formal GTM task, catalog-aware selection retrieves the evidence required for action, while the shared reader often fails to turn that evidence into a valid decision. The primary comparison remains uncertain; larger inputs and explicit supersession handling provide informative but bounded

results. The practical contribution is an auditable fixture suite, retained model evidence and a clear failure boundary: sufficient memory is not sufficient decision support, and explicit eligibility checks can outperform generative decisions when the rules are already known.

References

[1] Lorenzo Jaime Yu Flores, Junyi Shen, and Goodman Gu. Towards Reliable Multi-Agent Systems for Marketing Applications via Reflection, Memory, and Planning, 2025. URL https://arxiv.org/abs/2508.11120v2. arXiv preprint; consulted version.

[2] Xiang Shen, Yuhang Zhou, Yifan Wu, Zhuokai Zhao, Siyu Lin, Lei Huang, Qianqian Zhong, Lizhu Zhang, Benyu Zhang, Xiangjun Fan, and Hong Yan. Agentic Recommender System with Hierarchical Belief-State Memory, 2026. URL https://arxiv.org/abs/2605.14401v2. arXiv preprint; consulted version.

[3] Changyu Du, Alexander Vosseler, Filippo Mazza, and André Borrmann. IFCMemoryBench: Evaluating Long-Term Memory of LLM-Based Agents in BIM Information Retrieval, 2026. URL https://arxiv.org/abs/2607.26072v1. arXiv preprint; consulted version.

[4] Julia Belikova, Rauf Parchiev, Evgeny Egorov, Grigorii Davydenko, Gleb Gusev, Andrey Savchenko, and Maksim Makarenko. Managing Procedural Memory in LLM Agents: Control, Adaptation, and Evaluation, 2026. URL https://arxiv.org/abs/2606.23127v1. arXiv preprint; consulted version.

[5] Deepanjan Bhol. Sales Research Agent and Sales Research Bench, 2025. URL https://arxiv.org/abs/2602.17017v1. arXiv preprint; consulted version.

[6] Kung-Hsiang Huang, Akshara Prabhakar, Onkar Thorat, Divyansh Agarwal, Prafulla Kumar Choubey, Yixin Mao, Silvio Savarese, Caiming Xiong, and Chien-Sheng Wu. CRMArena-Pro: Holistic Assessment of LLM Agents Across Diverse Business Scenarios and Interactions, 2025. URL https://arxiv.org/abs/2505.18878v1. arXiv preprint; consulted version.

[7] Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory, 2024. URL https://arxiv.org/abs/2410.10813v2. arXiv preprint; consulted version.

[8] Di Wu, Zixiang Ji, Asmi Kawatkar, Bryan Kwan, Jia-Chen Gu, Nanyun Peng, and Kai-Wei Chang. LongMemEval-V2: Evaluating Long-Term Agent Memory Toward Experienced Colleagues, 2026. URL https://arxiv.org/abs/2605.12493v1. arXiv preprint; consulted version.

[9] Preston Rasmussen, Pavlo Paliychuk, Travis Beauvais, Jack Ryan, and Daniel Chalef. Zep: A Temporal Knowledge Graph Architecture for Agent Memory, 2025. URL https://arxiv.org/abs/2501.13956v1. arXiv preprint; consulted version.

[10] Chris Latimer, Nicoló Boschi, Andrew Neeser, Chris Bartholomew, Gaurav Srivastava, Xuan Wang, and Naren Ramakrishnan. Hindsight is 20/20: Building Agent Memory that Retains, Recalls, and Reflects, 2025. URL https://arxiv.org/abs/2512.12818v1. arXiv preprint; consulted version.

[11] Mihir Shriniwas Arya. RECON: Benchmarking Agent Memory for Compositional Reasoning over Long Contexts, 2026. URL https://arxiv.org/abs/2607.16716v1. arXiv preprint; consulted version.

[12] Jeongmin Bae, Yongjae Kim, Kyoung Hur, Donghyoung Han, and Min-Soo Kim. AkasicMEM: Governed Enterprise Memory for Agents, 2026. URL https://arxiv.org/abs/2609.25563v1. arXiv preprint; consulted version.

[13] Yixi Zhou, Sikun Wang, Lei Fan, and Fan Zhang. EviGraph: Proof-Carrying Selective Recommendation over Temporal Public-Service Knowledge Graphs, 2026. URL https://arxiv.org/abs/2610.00212v1. arXiv preprint; consulted version.

[14] Sen Gao, Yongan Li, Han Chen, Bingsheng He, Ming Wu, Jason Zeng, Michael Heinrich, and Shixuan Sun. MemDerive: A Diagnostic Benchmark for Agent Memory over Evolving Data, 2026. URL https://arxiv.org/abs/2610.25373v1. arXiv preprint; consulted version.

[15] Shangze Li, Yanyu Yao, Derong Xu, Zhi Zheng, and Tong Xu. From Memory Hits to Answerable Evidence: Query-Conditioned Memory Compilation for Long-Term Agent Memory, 2026. URL https://arxiv.org/abs/2610.23313v1. arXiv preprint; consulted version.

[16] Yongjie Zhou, Shuai Wang, Bevan Koopman, and Guido Zuccon. Do Conversational Agents Need Explicit Memory? A Controlled Comparison with Raw-History Retrieval, 2026. URL https://arxiv.org/abs/2610.24671v1. arXiv preprint; consulted version.

[17] Ruizhe Li, Licheng Zhang, Benfeng Xu, Mingxuan Du, Zheren Fu, and Weidong Chen. When Your Agent Opens the Chat App: Agent-Controlled Search over Raw Chat Logs Rivals Structured Memory, 2026. URL https://arxiv.org/abs/2608.12888v2. arXiv preprint; consulted version.

[18] Stephen Robertson and Hugo Zaragoza. The Probabilistic Relevance Framework: BM25 and Beyond. *Foundations and Trends® in Information Retrieval*, 4(1-2):1–174, 2009. doi: 10.1561/1500000019. URL https://doi.org/10.1561/1500000019.

[19] Gordon V. Cormack, Charles L A Clarke, and Stefan Buettcher. Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *Proceedings of the 32nd international ACM SIGIR conference on Research and development in information retrieval*, pages 758–759, 2009. doi: 10.1145/1571941.1572114. URL https://doi.org/10.1145/1571941.1572114.

[20] Zexue He, Yu Wang, Churan Zhi, Yuanzhe Hu, Tzu-Ping Chen, Lang Yin, Ze Chen, Tong Arthur Wu, Siru Ouyang, Zihan Wang, Jiaxin Pei, Julian Mcauley, Yejin Choi, and Alex Pentland. Benchmarking Agent Memory in Interdependent Multi-Session Agentic Tasks. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pages 41975–42005. PMLR, 2026. URL https://proceedings.mlr.press/v306/he26am.html. Consulted arXiv:2602.16313v2; proceedings record verified.

[21] Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. Mem0: Building Production-Ready AI Agents with Scalable Long-Term Memory, 2025. URL https://arxiv.org/abs/2504.19413v1. arXiv preprint; consulted version.

[22] Ruhan Wang, Kishan Panaganti, and Dongruo Zhou. More Memory, Worse Agents: Error Reproduction and Anti-Persistence in LLM Agents, 2026. URL https://arxiv.org/abs/2610.27406v1. arXiv preprint; consulted version.

[23] Yanzhao Zhang, Mingxin Li, Dingkun Long, Xin Zhang, Huan Lin, Baosong Yang, Pengjun Xie, An Yang, Dayiheng Liu, Junyang Lin, Fei Huang, and Jingren Zhou. Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models, 2025. URL https://arxiv.org/abs/2506.05176v3. arXiv preprint; consulted version.

[24] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 Technical Report, 2025. URL https://arxiv.org/abs/2505.09388v1. arXiv preprint; consulted version.

A Condition diagnostics and recorded failures

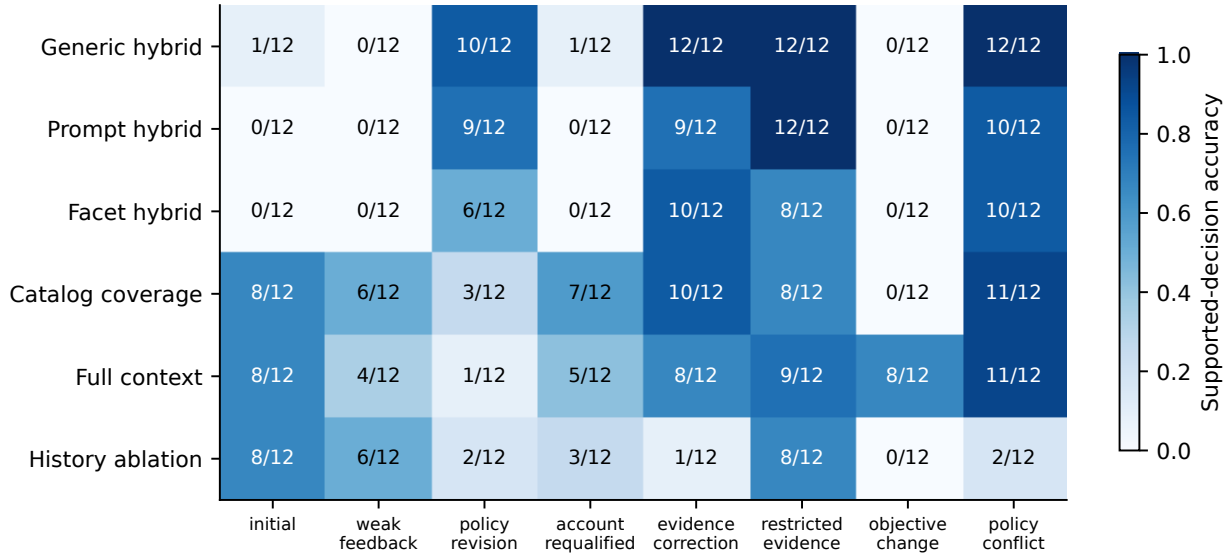


Figure 3: Primary correct decisions out of 12 worlds per condition. Columns follow protocol order, not one uninterrupted timeline. Initial, weak-feedback, account-requalified and objective-change cases are positive; the other four require withholding. Each cell uses the same worlds, so the cells are not independent samples.

The error-localization script selects the first lexicographic case in each category, preserving all other examples and paired disagreements. The following inspections use those pointers. They are research-assistant readings of recorded traces, not blinded expert assessment or representative sampling.

Withholding despite complete support. For `q0b2e9ee8ab`, catalog-coverage attempt 167 binds the approved seller role/geography and observed account role/geography correctly. Its

Method	Decision	Positive	Neg. rec.	Tokens	Gen. sec.
Prompt hybrid	43.8%	27.1%	39.6%	1523	53.9
Catalog coverage	61.5%	62.5%	39.6%	1519	52.5

Table 3: Exploratory 1,536-token comparison, with the same definitions and 96/48/48 denominators as Table 1. These queries reuse the same 12 worlds, not a new independent sample.

qualification alternative has no additional requirements. The output nevertheless requests partner status, channel fit, offering pain and procurement stage. Full-context attempt 211 instead selects the valid qualification alternative. This records failure to scope missing information to the requested alternative, despite the clarified common instructions.

Inapplicable recommendation with correct source bindings. For `q0d8088c402`, catalog-coverage attempt 516 recommends template P1. The supplied approved seller geography is UK, while the account geography is EU; the template also costs three units against a two-unit allowance. Its prerequisite table points to the actual supplied records. Correctly naming records therefore does not establish that their values satisfy equality or resource constraints. The scorer records an inapplicable template and prevents mock execution.

Proposal mistaken for approved policy. For `q1c366e6ee1`, catalog-coverage attempt 226 binds the approved-geography slot to S9, an unapproved CA proposal, instead of S2, the approved UK policy matching the account. The raw output and original bindings remain intact. The scorer rejects misbound, incomplete and irrelevant support; it does not silently substitute S2. This error illustrates why authority metadata must be checked in the decision, even when it is preserved in memory.

B Reproduction and experimental boundary

The companion separates four commands: offline contract checks, actual model inference, analysis of saved outputs, and manuscript compilation. The first requires Python 3.12; figure generation adds pinned NumPy and Matplotlib dependencies. Fresh inference requires separately downloaded hash-checked weights, the pinned `llama.cpp` source, compatible CPU instructions and approximately 32 GiB of memory. The main weights occupy about 19.2 GB including the embedding model. Model files, third-party papers and private product material are excluded from the companion. Preparing the package does not itself publish it or submit the report to a venue.

The generation checkpoint is Qwen3-30B-A3B-Instruct-2507, with approximately 30.5 billion total and 3.3 billion active parameters, in Q4_K_M quantization. We use the checkpoint's non-thinking mode rather than a generated reasoning trace. The embedding checkpoint is Qwen3-Embedding-0.6B in Q8_0, using its final/EOS representation, 1,024 dimensions and normalized cosine similarity [23, 24]. Quantization and deployment choices are resource constraints, not model-compression contributions. Complete upstream revisions, SHA-256 hashes, server flags and binary hashes are saved with the runs.

The reader serves a 16,384-token context with one slot, four CPU threads and prompt caching disabled; the embedding service uses a separate loopback endpoint. Each planned decision has temperature 0, seed 20261022 and at most 280 output tokens. The request body, schema field order and all returned usage are retained. Other sampler defaults are recorded from the running

server, rather than inferred from an external API's defaults. A fixed seed is not a guarantee of cross-hardware bitwise determinism.

The primary block orders its 576 attempts with seed 20261022; the secondary block orders 192 attempts with seed 20261023. There is no generation-response reuse between blocks. The secondary block can reuse the common embedding index, retaining computation origins. A transport failure stops execution for repair. Resumption preserves completed and failed attempts, marks issued attempts without terminal records as interrupted failures, and starts only previously unissued work. An exclusive lock prevents two processes from executing one run concurrently. Unknown usage from an interruption remains unknown rather than being assigned zero cost.

The provenance audit reconstructs prompts from the original eligible facts and public catalog, checks source aliases and output grammar, and compares exact serialized request settings. Offline mutation checks reject rehashed forged facts, altered decoding settings and duplicate physical-call records. These checks support artifact consistency; they are neither independent expert validation of the task nor a formal security proof.

The primary interval uses 10,000 bootstrap draws with seed 7201, resampling four worlds with replacement within each of the three held-out families. The effect is the average of paired world differences. Other comparisons and the input-budget sensitivity explain behavior; they are not a multiplicity-adjusted family of superiority tests. The public freeze files record this analysis specification, the 13 source-file hashes and both dataset hashes before main inference.

A descriptive error-localization report was added after the freeze but before main quality outcomes were inspected. It crosses complete-witness availability with positive decision correctness and preserves the original scores and denominators. Exclusive categories distinguish execution/output failures, missing retrieved support, withholding despite available support, inapplicable choices, and invalid bindings. All paired disagreements are retained; illustrative trace pointers select the first lexicographic case within each category. These summaries support inspection, not causal mediation estimates or additional confirmatory tests.

An input-only audit added after the freeze, before main outcome inspection, constructs one complete minimal-slot witness for each of the 48 answerable queries. All fit 1,024 tokens: the largest requires 882 with the generic prompt and 951 with the checklist. This establishes representational capacity, not retrieval or reader success, token-optimal selection or a new evaluated method. Messages, rendered prompts, tokenizer receipts and hashes are retained in `analysis/diagnostics/input-capacity.json`. No generation call or experimental outcome enters this check.

A corresponding pre-results output audit constructs one correct decision per query from the public contract. All 96 fit the 280-token allowance: compact JSON requires at most 127 tokens and two-space indentation at most 170, including one end token. This does not establish successful generation or correct explanations. Exact strings, tokenizer receipts and offline checks are in `analysis/diagnostics/output-capacity.json`; the diagnostic reads neither labels nor experimental responses.

C Fixture semantics and development disclosure

The main generator seed is 2026102202; the development seed is 2026102201. The main families are renewal, partner and compliance. Their task-specific relations respectively connect customer state and usage trend; partner status and channel fit; or certification and an observed compliance signal to an offering pain and proof. These literal requirements are public catalog definitions. Their presence is not a hidden reward function or a learned estimate of commercial utility.

Every world has four answerable and four unanswerable queries. Approved policy revision

initially creates a targeting mismatch; a later account observation restores fit. An observation learned still later corrects an earlier signal, even when the valid-time cutoff is held fixed. Restricted proof is unavailable to a requester lacking its required scope. An unrelated approved role without a supersession link creates a genuine conflict. Weak feedback and an unapproved policy proposal do not themselves change approved strategy. The positive changed-objective case requests only a qualification audit. Labels and condition names remain outside reader messages.

The two main research alternatives share prerequisites. The three-step variant researches the account, verifies prerequisites and prepares a report; the two-step variant omits the first step. A world-specific allowance of two or three artificial units controls their eligibility. A separate procurement alternative requires an executive-only source. The main histories contain 28–56 events each. There are three worlds at each distractor count of 12, 20, 28 and 36. Zero, one or two duplicate pairs occur in three, five and four worlds respectively, sampled independently of family. Duplicate records carry the same value and do not create additional facts. These realized counts are saved in the dataset's `design-summary.json`.

Development failures are retained in the companion's tuning record and raw ledgers. The first compact citation interface proved unreliable with the smaller readers; a schema-serialization defect was also corrected before the main freeze. Subsequent common-interface changes replaced long identifiers with exact source labels, then required a prerequisite table and clarified that a missing prerequisite blocks only alternatives that require it. No missing source is filled by the evaluator. The final 48-call pilot established that even full context could fail despite containing a complete witness. Main evaluation therefore measures both selection and utilization, rather than assuming that retrieval availability implies a correct recommendation.

For the final pilot, a diagnostic taxonomy correction changed repeated source bindings from a flat-citation shape error to a per-slot semantic error. All 48 primary correctness outcomes remained unchanged. The original records and the audit are preserved. This correction does not remove wrong bindings, repair model outputs or alter the main task after observing its results. The final frozen materializer deduplicates citation sets while preserving every original prerequisite binding for validation.

D Authorship, data and release status

This AI-assisted report was researched, implemented and drafted in a cloud session under the user's direction. It has no independent human annotation, external review or peer-reviewed acceptance. Cited results remain source-reported unless explicitly reproduced. The companion licenses original fictional data and saved outputs separately from code; it requires no customer information or private production source. Public release and paper submission are separate milestones.